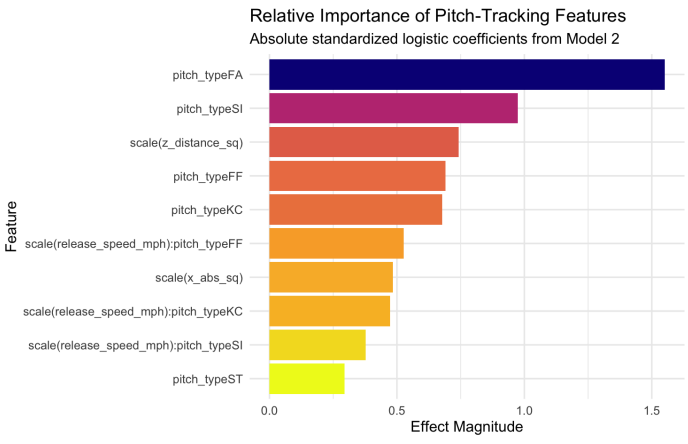
Pitch Outcome Model

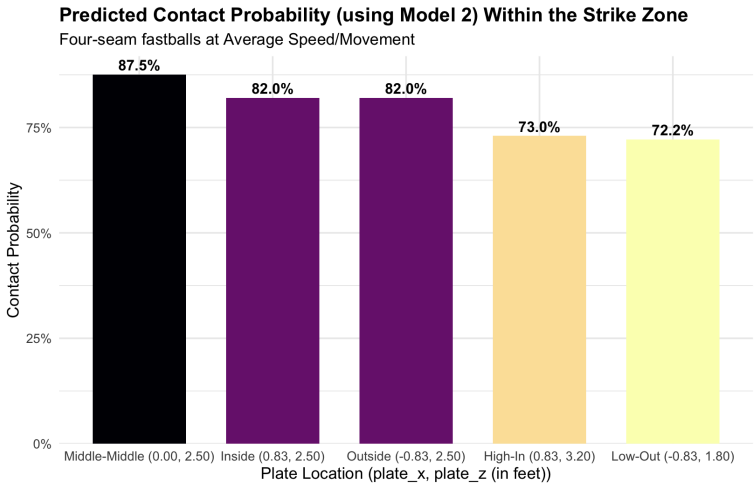
In this project, I developed models to estimate the probability that a batter makes contact with a pitch, given that the batter swung at the pitch. Using pitch tracking data on location, velocity, and movement, the goal was to connect measurable pitch traits to swing outcomes and evaluate which pitchers limit contact beyond what their pitch characteristics would predict. The analysis had three stages: first, building an improved logistic regression model with additional features derived from pitch trajectory; second, extending the analysis with a gradient boosting model to test whether a nonlinear approach improved predictive accuracy; and third, validating both models on held-out data using negative log loss before using the best-performing model to compare each pitcher's expected and observed contact rates.

Part 1: Predictive Modeling

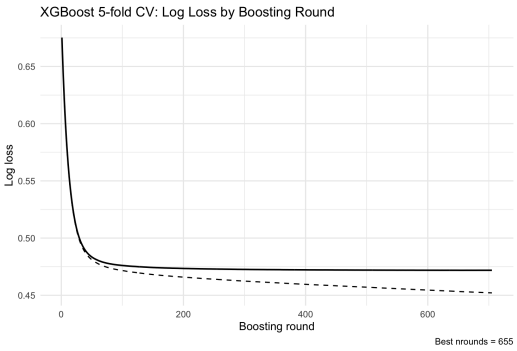
To improve upon the simple logistic regression introduced in class, I added new predictor variables derived from pitch tracking data to better capture the many different characteristics of a pitch. In addition to horizontal and vertical location, the model included the horizontal distance from the center of home plate and the vertical distance from the midpoint of the strike zone, along with squared terms for both. These squared features allow contact probability to drop more sharply as a pitch moves away from "middle-middle," reflecting how even small deviations in location can affect contact probability greatly. I also added physical characteristics such as release speed, extension, induced vertical break, and horizontal break to represent how velocity and movement interact to influence contact. To capture game context, I introduced "count_pressure," which measures the difference in expected run value between a potential strike and a potential ball based on the current count. Categorical pitch type was added to capture the nuances of different pitch families without modeling every individual pitch characteristic, and an interaction between velocity and pitch type allowed the model to capture how velocity influences contact within each pitch type. The resulting feature importance plot, which you can see to the left, shows that location is one of the strongest drivers of contact probability, with "z_distance_sq" and "x_abs" ranking highly in the plot. This aligns with



baseball intuition, as pitches along the inside/outside edges or on the corners are much harder to make contact with, regardless of speed or movement. The bar chart to the right reinforces this point, showing that predicted contact probability peaks in the center of the zone and decreases near the edges and corners of the zone. When the model was first fit, the rankings were dominated by rare pitch types such as knuckleballs, screwballs, and eephuses.



Because these appear so infrequently, their coefficients were estimated from very few observations, which exaggerated their influence. After filtering out those outliers, the results became much more stable and interpretable. The biggest remaining weakness of the model is that even with interaction and squared terms, logistic regression can only approximate nonlinear relationships among pitch traits—such as how horizontal and vertical movement interact differently by pitch location or count. To address this, I trained a gradient-boosting model with XGBoost, which naturally captures these complex effects. The model was tuned through five-fold cross-validation, testing combinations of learning rate, tree depth, and regularization parameters to minimize test log loss. Performance improved steadily before plateauing around 655 boosting rounds, where the mean test log loss reached 0.4736, indicating a well-fit model without overfitting. The final tuning parameters—a learning rate of 0.05, tree depth of 5, minimum child weight of 5, subsample of 0.9, and column sample of 0.8—balanced flexibility and stability, allowing the model to capture nonlinear pitch patterns while maintaining generalization.



Part 2: Model Validation

To evaluate how well each model generalized to unseen data, I estimated out-of-sample error using negative log loss, which measures how well predicted probabilities align with actual outcomes. For the logistic regressions, I randomly split the dataset into a training set (75%) and test set (25%), fit the models on the training data, and computed log loss on the held-out set. The

metric penalizes confident errors more heavily, using -log(p) when contact occurs and -log(1-p) when it does not. Averaging these values across all pitches gives the model's overall log loss, where lower values indicate better calibration. The baseline logistic model, which only used count, pitch location, and velocity, achieved a test log loss of 0.4936. After adding location-derived, movement, and contextual features, the improved model reduced the test log loss to 0.4866, showing a small but meaningful improvement. The XGBoost model, trained with five-fold cross-validation to minimize log loss, performed best at 0.4736, confirming that its nonlinear structure captured interactions among movement, velocity, and location more effectively than logistic regression.

Part 3: Player Evaluation

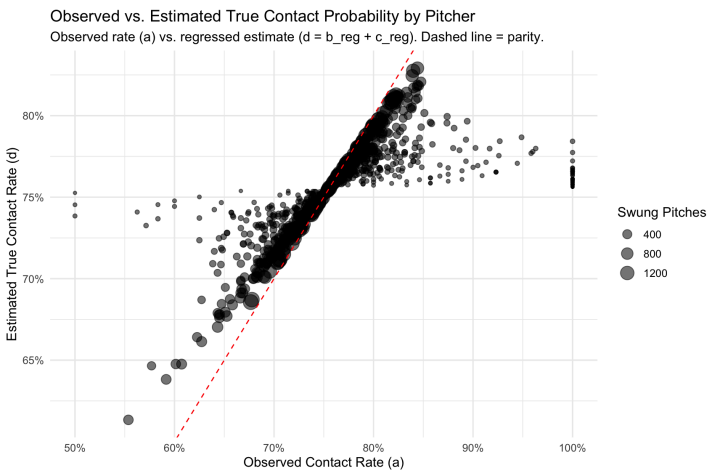
Pitcher Leaderboard — Lowest Predicted Contact Probability (d_pred)

Regressed-to-mean estimates using Model 2

Rank	Pitcher	ID	Swung Pitches	Observed (a)	Expected (b)	Residual (c)	Reg. (c)	True Contact (d)
1	Mason Miller	695243	475	55.4%	64.4%	-9.0%	-6.2%	58.1%
2	Fernando Cruz	518585	312	57.7%	66.4%	-8.7%	-5.2%	61.2%
3	Edwin Díaz	621242	502	59.2%	67.7%	-8.5%	-6.0%	61.7%
4	Alex Lange	656638	12	58.3%	62.8%	-4.5%	-0.2%	62.6%
5	Brendon Little	663893	464	60.1%	69.1%	-9.0%	-6.2%	62.9%
6	Bryan Abreu	650556	532	60.7%	69.4%	-8.7%	-6.2%	63.2%
7	Jack Little	669193	25	68.0%	63.8%	4.2%	0.5%	64.2%
8	Josh Hader	623352	440	62.3%	71.4%	-9.1%	-6.2%	65.2%
9	Devin Williams	642207	549	62.8%	71.7%	-8.9%	-6.4%	65.3%
10	Ronny Henriquez	678692	553	65.8%	65.6%	0.3%	0.3%	65.8%
11	Jeremiah Estrada	669093	636	65.3%	67.5%	-2.3%	-1.6%	65.9%
12	Andrés Muñoz	662253	471	64.5%	69.1%	-4.5%	-3.1%	66.0%
13	Mason Montgomery	682254	427	65.6%	67.7%	-2.1%	-1.4%	66.3%
14	Griffin Jax	643377	533	65.1%	70.3%	-5.2%	-3.7%	66.6%
15	Connor Phillips	683175	150	64.7%	68.0%	-3.3%	-1.4%	66.6%
16	Edwin Uceta	670955	628	64.3%	73.4%	-9.1%	-6.8%	66.6%
17	Nick Sandlin	680704	130	63.8%	68.4%	-4.6%	-1.7%	66.7%
18	Aroldis Chapman	547973	513	64.5%	72.5%	-8.0%	-5.7%	66.9%
19	Ryan Burr	621114	14	50.0%	68.1%	-18.1%	-1.2%	66.9%
20	John Rooney	680701	8	50.0%	67.6%	-17.6%	-0.7%	66.9%

To evaluate pitcher performance, I used the improved logistic regression model to predict contact probability for each swung pitch and averaged those predictions by pitcher. I then compared each pitcher's observed contact rate with their expected rate, and inputted those residuals into a linear mixed effects regression model, which shrinks each pitcher's residual toward the league mean based on sample size. The resulting predicted true contact rate provides a more stable estimate of how much contact a pitcher truly allows.

The leaderboard of pitchers with the lowest predicted true contact rates is dominated by elite relievers such as Mason Miller, Edwin Díaz, Bryan Abreu, and Devin Williams, all of whom pair upper-90s fastballs with elite sliders (with the exception of Williams, who has an elite changeup instead). From our leaderboard, we can see that for the most part, pitchers who combine elite velocity with a secondary pitch that moves sharply and tunnels well off of their fastball tend to suppress contact most effectively. It is also interesting that every single member of the top 20 is a reliever, and there are only 3 starters in the top 40: Blake Snell (25th), Dylan Cease (32nd), and Tarik Skubal



(37th). All three possess some of the best fastballs among starting pitchers, supported by an elite changeup in the cases of Snell and Skubal, and an elite slider in Cease's case.

The scatterplot comparing observed and estimated true contact rate shows a strong linear relationship along the parity line, indicating that the model captures pitcher outcomes well and exhibits no clear bias. Only a few points fall below an estimated contact rate of 70%, representing those elite strikeout pitchers with exceptional swing-and-miss ability. There are many pitchers with smaller sample sizes that have observed contact rates above 80% but lower true contact rates, suggesting their results were somewhat unlucky within a small sample—despite allowing frequent contact, their pitch traits, such as movement and command, should yield better outcomes over time.

Two pitchers who stood out on opposite ends of the model were Matt Bowman, who allowed more contact than expected, and Erik Sabrowski, who allowed less. Bowman's underperformance stems from an east-west movement profile with little vertical separation, reducing tunneling and making his arsenal easier to track visually for the hitter. Combined with his 10th-percentile fastball velocity and standard 30-degree arm angle, his delivery offers limited deception, helping explain his higher observed contact rate. Sabrowski, by contrast, overperformed model expectations. His higher 48-degree arm angle adds deception, and his three distinct pitch shapes—a fastball with elite ride, a curveball with steep drop, and a gyro slider bridging the two—create strong vertical (and slight horizontal) separation and tunneling. The model likely underrates this combination, as it cannot fully capture visual deception or sequencing. Together, these cases show how factors like arm angle, tunneling, and pitch design can drive real differences in contact outcomes beyond what raw metrics alone explain.

Conclusion

This project demonstrated how statistical and machine learning models can quantify the factors that drive contact in baseball. The logistic regression identified key physical and contextual factors, while XGBoost better captured nonlinear relationships. Applying these models to pitcher evaluation revealed the general trend that pitchers with high velocity fastballs and one or two elite secondary pitches are the best at consistently suppressing contact. The analysis also showed how regression to the mean distinguishes sustainable skill from random variation, and we were also able to see that differences in deception, tunneling, and pitch design can explain why some pitchers outperform or underperform their expected results.